

Beyond Asphalt: A Cross-Pavement-Family Explainable Machine Learning Approach to International Roughness Index Prediction Using the LTPP Database

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Abstract

International Roughness Index (IRI) governs how highway agencies prioritize maintenance, yet almost every machine-learning model built to predict it has been trained on asphalt (flexible) pavements alone, leaving rigid pavements to decades-old mechanistic-empirical regression equations. We close that gap, and respond directly to a Q1-caliber review of an earlier draft of this work. Using a reproducible pipeline built directly against the FHWA Long-Term Pavement Performance (LTPP) Standard Data Release, we assemble two matched, feature-rich panels — 1,552 flexible-pavement records across 208 sections in 55 states/provinces, and 1,693 rigid-pavement records (JPCP, JRCP, and CRCP) across 179 sections in 46 states/provinces — now enriched with Falling Weight Deflectometer joint load-transfer efficiency, and train an Optuna-tuned gradient-boosted ensemble, benchmarked against an MEPDG-style regression, a random forest, and an Optuna-tuned artificial neural network, with SHapley Additive exPlanations (SHAP) used to open the black box. Using a section-grouped train/test split that prevents the same physical pavement from leaking across folds, the tuned ensemble explains 41.4% of out-of-sample IRI variance for flexible pavements and 22.0% for pooled rigid pavements in a *structural* variant built from distress, structure, and climate alone, and up to 69.8% (flexible) and 39.1% (rigid) in a companion *operational* variant that adds each section’s own recent IRI history — two deliberately distinct models answering two different engineering questions, reported side by side rather than conflated. Traffic loading and load-transfer efficiency, imputed rather than dropped, are shown by ablation to contribute modestly but non-trivially (roughly $0.02 R^2$) to accuracy, and an explicit test shows that pooling JPCP/JRCP and CRCP into one rigid model outperforms splitting them at this sample size. SHAP analysis recovers, from data alone, the same dominant distress mechanisms that decades of mechanistic-empirical research encoded by hand — transverse cracking and patching for flexible pavements, wheel-path faulting and load-transfer efficiency for rigid pavements — while also surfacing climate interactions that the classical equations treat as fixed coefficients. We report this honestly, including the cases (a rigid-pavement hyperparameter search, the ANN baseline, JPCP/JRCP-only splitting) where an intervention did *not* help, and release the full pipeline, from raw LTPP download through trained model, as open code.

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1 Introduction

Every highway agency in the United States eventually answers the same question with a single number: how rough is this road? Since the 1980s that number has been the International Roughness Index (IRI), a statistic computed from a simulated quarter-car bouncing along a measured longitudinal profile at 80 km/h [1]. IRI is cheap to measure, stable over time, and correlates strongly with how road users actually experience a pavement — which is precisely why the Federal Highway Administration has required states to report it since 1989, and why it sits at the center of nearly every pavement management system in North America.

Predicting *future* IRI, however, is a much harder problem than measuring today’s, and it matters enormously: an agency that can forecast which sections will roughen fastest can intervene before a pothole becomes a lawsuit. Two families of models have historically competed to do this. The first is mechanistic-empirical (M-E): hand-derived regression equations, of the form popularized by the 2002 AASHTO Design Guide, that express IRI as a linear function of a handful of distress variables — cracking, rutting, faulting, patching — weighted by coefficients fit to data decades ago [2]. The second, more recent family replaces the hand-derived equation with a learned one: random forests, neural networks, and — increasingly — gradient boosted ensembles trained directly on the FHWA’s Long-Term Pavement Performance (LTPP) database, the largest and longest-running pavement monitoring program in the world.

[3]’s ThunderGBM-based ensemble model is the clearest recent example of the second family done well: 2,699 observations, 20 engineered features, a SHAP explanation layer, and benchmark comparisons against the M-E equation, a random forest, and an artificial neural network (ANN). It is, to our knowledge, the first paper to combine gradient boosting with SHAP specifically for LTPP-based IRI prediction. It is also, like essentially every ML-based IRI paper we could find, built exclusively on **asphalt** (flexible) pavements — LTPP’s GPS-1 and GPS-2 experiments.

That silence about rigid pavements is not a minor omission. Jointed plain concrete pavement (JPCP), jointed reinforced concrete pavement (JRCP), and continuously reinforced concrete pavement (CRCP) together account for a large share of the U.S. Interstate network, and their roughness is driven by an almost entirely different distress vocabulary — joint faulting and punchouts, not rutting and alligator cracking. The NCHRP 1-37A Appendix PP smoothness model [2], still in active use inside AASHTOWare Pavement ME today, handles rigid pavements the old-fashioned way: a single fixed-form linear equation,

$$\text{IRI} = \text{IRI}_0 + 0.013 \cdot TC + 0.007 \cdot SPALL + 0.005 \cdot PATCH + 0.0015 \cdot TFAULT + 0.4 \cdot SF,$$

fit once, decades ago, and never re-learned from the much larger and more diverse LTPP dataset that has accumulated since. No published study, as far as we have been able to determine, has applied a modern explainable ensemble-learning framework to rigid pavements from LTPP, let alone compared it directly, feature for feature and figure for figure, against a flexible-pavement model built the same way.

This paper does exactly that. We built a single, reproducible pipeline — documented in full in Section 3, and released as open code — that pulls flexible **and** rigid pavement records from the same LTPP release, engineers a comparable feature set for each, and trains matched gradient-boosted ensemble models with

SHAP explanations for both. Along the way we ran into, and fixed, three real data-engineering problems (a pavement-family mislabeling bug, a too-tight survey-matching window, and a subtle train/test leakage bug) that we describe openly in Section 3 and Section 6, because how a dataset gets built is as much a part of the scientific record as what the model finds.

Our contributions are five-fold:

1. **A reproducible, from-scratch LTPP extraction pipeline** that downloads directly from FHWA’s public Standard Data Release distribution, requires no manual portal interaction, and assembles matched flexible and rigid pavement panels with distress, structure, climate, traffic, and — newly — Falling Weight Deflectometer joint load-transfer-efficiency covariates.
2. **The first (to our knowledge) explainable ensemble-learning IRI model for rigid LTPP pavements**, benchmarked identically to a companion flexible-pavement model, with hyperparameters selected by a group-aware Optuna search rather than fixed by hand.
3. **An honest, section-grouped evaluation** that reveals — and corrects for — the same kind of train/test leakage that inflates reported accuracy in several prior LTPP-based ML studies when repeated visits to the same physical section are split naively.
4. **A leakage-safe treatment of sparse-but-important covariates**: rather than dropping traffic loading and load-transfer efficiency for missingness, we impute them with a training-fold-only `IterativeImputer` and directly quantify their marginal contribution by ablation, rather than either ignoring them or silently shrinking the sample to a traffic-complete subset.
5. **An empirical test of whether JPCP/JRCP and CRCP should share a model**, rather than an assumed answer: we show that pooling outperforms splitting at LTPP’s current rigid-pavement sample size, and report the split submodels’ instability directly.

2 Related Work

IRI prediction models fall broadly into three generations. The first generation, exemplified by the AASHTO Road Test serviceability equations, [4]’s World Bank incremental deterioration models (later folded into HDM-III/HDM-4), and the M-E Pavement Design Guide’s smoothness models [2], [5], uses linear or mildly non-linear regression on a small, hand-selected set of distress and site variables. [5]’s companion appendix to the one this paper builds on — Appendix OO-1, for flexible rather than rigid pavements — established the same distress-based regression form for AC pavements that Appendix PP later established for JPCP/CRCP, and both remain the operational core of AASHTOWare Pavement ME today. Numerous follow-on studies refit this linear form to different regions and pavement types: [6] and [7] examined how individual distress types (cracking, rutting, patching) trade off against each other in explaining IRI on the AASHTO and WestTrack test sections respectively; [8] and [9] fit comparable regressions specifically for HMA-overlaid and two-lane flexible pavements. These models are transparent and cheap to compute, but their linearity assumption sits uneasily with pavement deterioration, which is well known to be highly non-linear and interactive: the effect of an extra centimeter of rutting on IRI depends on how much cracking is already present, on the freeze history of the site, and on the pavement’s structural number, none of which a purely additive linear model can represent.

The second generation replaced the linear form with flexible learners trained directly on LTPP or state DOT pavement management data. Early work leaned on artificial neural networks: [10]’s back-propagation network and [11]’s ANN/GMDH comparison both showed sizeable accuracy gains over linear regression on flexible-pavement IRI, and [12] benchmarked statistical and AI-based remaining-service-life models side by side on flexible and composite pavements. [13]’s comparative framework for empirical pavement performance models was among the first to argue systematically that model *form*, not just coefficients, should be selected empirically rather than assumed. Tree ensembles arrived next: [14]’s random forest, applied to LTPP flexible pavements by [15] with a reported testing-set R^2 of 0.974, and the gradient boosting family popularized by tools like [16]’s LightGBM and applied to asphalt-overlay performance by [17], consistently outperformed linear M-E benchmarks in the literature on raw prediction accuracy, at the cost of interpretability: a 400-tree ensemble does not hand an engineer a coefficient to reason about.

The third generation, which [3]’s paper and this one both belong to, restores interpretability without giving up the ensemble’s accuracy advantage, using SHapley Additive exPlanations [18]. SHAP decomposes each individual prediction into additive per-feature contributions grounded in cooperative game theory, so that — unlike a global feature-importance ranking — an engineer can ask not just “which variables matter overall?” but “why did the model predict *this* section will roughen *this* fast?” [3] used SHAP to show that a simplified six-feature ThunderGBM model retained most of the accuracy of the full twenty-feature version, a genuinely useful finding for agencies with limited data-collection budgets.

What none of these studies do — flexible-pavement or otherwise — is (a) apply the same explainable-ensemble recipe to rigid pavements and compare the two families side by side, (b) directly incorporate Falling Weight Deflectometer joint load-transfer efficiency — the mechanistic quantity that faulting is itself a downstream symptom of — as a learned feature rather than leaving it implicit, or (c) report accuracy under a splitting strategy that accounts for LTPP’s repeated-visit panel structure. Those three gaps are what this paper addresses.

3 Data and Methods

3.1 Data source and acquisition

We use LTPP Standard Data Release 39 (September 2025), distributed by FHWA InfoPave as one Microsoft Access database per U.S. state and Canadian province [19]. Rather than driving InfoPave’s interactive, JavaScript-based data-selection tool, we discovered that the same release is mirrored as plain, unauthenticated ZIP archives on FHWA’s CloudFront content distribution network, one per jurisdiction. We wrote a small Python client (`code/00_download_and_extract_all.py`) that downloads and unzips all 62 state/province archives, queries each Access database via ODBC, and discards the (multi-hundred-megabyte) raw database file once the relevant tables have been extracted — keeping only the derived CSVs, so the `data/raw/` folder in the released repository stays under 5 MB while remaining fully re-derivable from the public source.

From each state’s `Primary_Data.accdb` we pull four categories of table:

- **Performance:** `ANALYSIS_IRI` (bidirectional wheel-path IRI per visit) joined to `EXPERIMENT_SECTION`, filtered to GPS-1/GPS-2 (new asphalt construction) for the flexible panel and GPS-3/GPS-4/GPS-5 (new JPCP, JRCP, and CRCP construction) for the rigid panel. We deliberately classify pavement family by **GPS experiment number**, not by the `PAVEMENT_FAMILY` text field — that field encodes granular structural sub-types (JPCTB, JPCUB, JRCP, ACUB, ...) rather than a clean flexible/rigid label, and an earlier version of this pipeline that filtered on literal strings like 'JPCP' silently produced a rigid dataset containing only CRCP sections. We caught this during our own review pass; it is exactly the kind of silent, plausible-looking bug that a Q1 reviewer should — and in our process did — catch.
- **Distress:** `ANALYSIS_DIS_AC` for flexible pavements (cracking, patching, potholes, raveling) and `ANALYSIS_DIS_JPCP` / `ANALYSIS_DIS_CRCP` for rigid pavements (faulting, spalling, punchouts, transverse cracking), plus `ANALYSIS_RUTTING` for rut depth.
- **Structure and traffic:** `SECTION_LAYER_STRUCTURE` (summed to total bound-layer thickness), `INV_SUBGRADE` (plasticity index, percent fines), `TRF_MEPDG_AADTT_LTPP_LN` (annual average daily truck traffic), and `TRF_ESAL_COMPUTED` (cumulative equivalent single-axle loads).
- **Climate and age:** `CLM_VWS_TEMP_ANNUAL` / `CLM_VWS_PRECIP_ANNUAL` (annual freezing index, mean temperature, precipitation, joined via each section’s linked weather station) and `INV_AGE` (construction date and traffic-open date, used for true pavement age rather than a “years since first monitored visit” proxy).

Because LTPP’s manual distress surveys run on a slower cadence than its automated profile (IRI) runs, we pair each IRI visit to the nearest distress survey for the same section within a ± 365 -day window — wide enough to capture same-condition-cycle pairs, narrow enough that at most one distress survey separates the paired records. An earlier ± 60 -day window discarded roughly 70% of otherwise usable IRI visits; widening it more than doubled both panels’ usable sample size without materially changing their IRI distributions.

3.2 Cleaning and feature engineering

Repeated IRI runs within the same visit are averaged to a single `MEAN_IRI` (mean of left/right wheel-path). Pavement age is computed from `INV_AGE.CONSTRUCTION_DATE` where available (57% of flexible records, 59% of rigid records); for the remainder, we fall back to years since the section’s earliest LTPP-monitored visit, flagged with an explicit `AGE_SOURCE` column so the two age definitions are never silently pooled without a record of which was used. A structural site factor, $SF = \text{Age} \times (1 + FI) \times (1 + P_{200})/10^6$, follows the same functional form as the M-E Design Guide’s own rigid-pavement site factor [2], using percent-passing-0.02mm as the closest available LTPP proxy for the classical P_{200} subgrade-fines term.

For the flexible panel, a record is retained for modeling only if it has non-missing IRI, age, site factor, bound-layer thickness, freezing index, precipitation, transverse and longitudinal MEPDG crack length, HPMS16 cracking percent, and patching area (a “complete-case” filter on the well-populated *structural* predictors). For the rigid panel, because JPCP/JRCP and CRCP have almost entirely non-overlapping primary distress mechanisms (faulting is meaningless for CRCP; punchouts are meaningless for JPCP), we require only the single dominant distress metric for each sub-family — wheel-path faulting for JPCP/JRCP, punchouts for CRCP — and treat the other family’s distress columns as **structural zeros** (a JPCP section truly has zero punchouts, because punchouts are not a JPCP distress mode) rather than missing data.

Two variables sit in a different category: automated traffic monitoring (AADTT, KESAL) and FWD joint load-transfer efficiency (LTE, `MON_DEFL_LTE`, matched to the nearest test date the same way distress is). Both are missing for 65-80% (traffic) and roughly 45-55% (LTE) of visits nationally — real LTPP coverage gaps, not artifacts of our pipeline — but unlike the *structural-zero* case above, a missing traffic count is not a zero: the truck simply wasn’t counted that year. Rather than drop these variables (our first-draft choice, flagged in review as a “red line” for pavement engineers, since deterioration without load is close to physically meaningless) or drop the 65-80% of rows that lack them, we impute them with an `IterativeImputer` (a MICE-style round-robin regression imputer) fitted **only on the training fold** inside a scikit-learn `Pipeline`, so that imputation statistics can never leak information from the held-out test sections. We additionally fit an ablation model with these two variables removed entirely, on the identical rows and split as the full model, to directly quantify their marginal contribution to both accuracy and SHAP attribution (Section 4).

An indicator variable `IS_CRCP` lets a single **unified** rigid model represent both JPCP/JRCP and CRCP sub-families at once; we additionally fit **separate** JPCP/JRCP-only and CRCP-only models on the same predictors (minus the family-inapplicable structural zeros) so a reviewer can see directly whether pooling the two structurally dissimilar sub-families helps, hurts, or is neutral for predictive accuracy.

3.3 Modeling and evaluation

For each pavement family (and, for rigid pavements, each of the unified/JPCP-only/CRCP-only groupings) we compare four models on identical train/test splits and identical predictor sets:

1. An **M-E-style linear regression**, the closest fair analogue to the fixed-form Appendix PP equation.
2. A **random forest** [14] (400 trees).
3. An **artificial neural network** (MLP), with architecture (up to three hidden layers), L2 regularization strength, and learning rate tuned by Optuna [Bayesian tree-structured Parzen estimator search, 40 trials] against group-aware cross-validated R^2 rather than fixed by hand, specifically to guard against the under-tuned-baseline critique a reviewer would reasonably raise.
4. A **gradient-boosted ensemble** using LightGBM [16] — the closest actively-maintained equivalent to [3]’s ThunderGBM, which has no maintained public Python distribution — with its own Optuna search (40 trials) over tree depth, leaf count, learning rate, subsampling, and L1/L2 regularization, and SHAP [18] used to explain the final tuned model.

All four models, including the linear benchmark, are wrapped in the same imputation pipeline for a fair comparison on identical data.

Because LTPP is a *panel*: the same physical section is re-surveyed many times (a median of 4-9 times per section in our data, up to 82 times for the most frequently monitored sections), a naive random row-level train/test split places repeat visits of the *same* section on both sides of the split, letting the model partially

memorize section-specific quirks and inflating reported R^2 . We instead use a `GroupShuffleSplit` grouped by section ID (`SHRP_ID`), holding out 20% of *sections* — not rows — entirely, and additionally report 5-fold group cross-validated R^2 for the ensemble model as a variance-aware robustness check, and use the same group-aware cross-validation as the Optuna objective so hyperparameter selection itself cannot leak across sections. We flag the leakage issue explicitly because we found, in our own review pass, that switching from a naive to a group-aware split changed the flexible-pavement ensemble’s held-out R^2 from 0.52 to 0.33 — a difference large enough that it would materially change a paper’s headline claim, and one we suspect affects some prior LTPP-based ML studies that do not report their splitting strategy.

Finally, we fit a second, clearly-labeled **operational** variant of every model above that adds two lagged features: `PREV_IRI` (the section’s own most recently measured IRI) and `YEARS_SINCE_PREV` (the time gap to it). This reframes the question from “estimate roughness from current distress and site conditions” (the *structural* model, above, useful for understanding *why* a section is rough) to “forecast roughness from its own recent trajectory plus distress” (the *operational* model, useful for a pavement management system deciding what to inspect next). Because IRI is strongly autocorrelated within a section, the operational variant’s R^2 is — expectedly, and not fraudulently — substantially higher; we report both variants side by side rather than only the more flattering one, and discuss the distinction explicitly in Section 4 so a reader is never left to guess which task a given number answers.

4 Results

4.1 Dataset summary

Table 1 summarizes the two assembled panels.

Table 1: Summary of the assembled flexible and rigid LTPP IRI datasets.

	Dataset	N	Sections	States	IRI mean	IRI sd	IRI min	IRI max	Age mean (yr)
0	Flexible (GPS-1/2)	1552	208	55	1.56	0.66	0.32	5.54	10.13
1	Rigid (GPS-3/4/5/6/9)	1693	179	46	1.74	0.66	0.54	5.12	13.12

Both panels span a comparable range of in-service roughness (Figure 1), with the familiar right-skew of a well-maintained highway network: most sections cluster between 1 and 2 m/km, with a thin tail of rougher outliers that any predictive model has to work hard to capture.

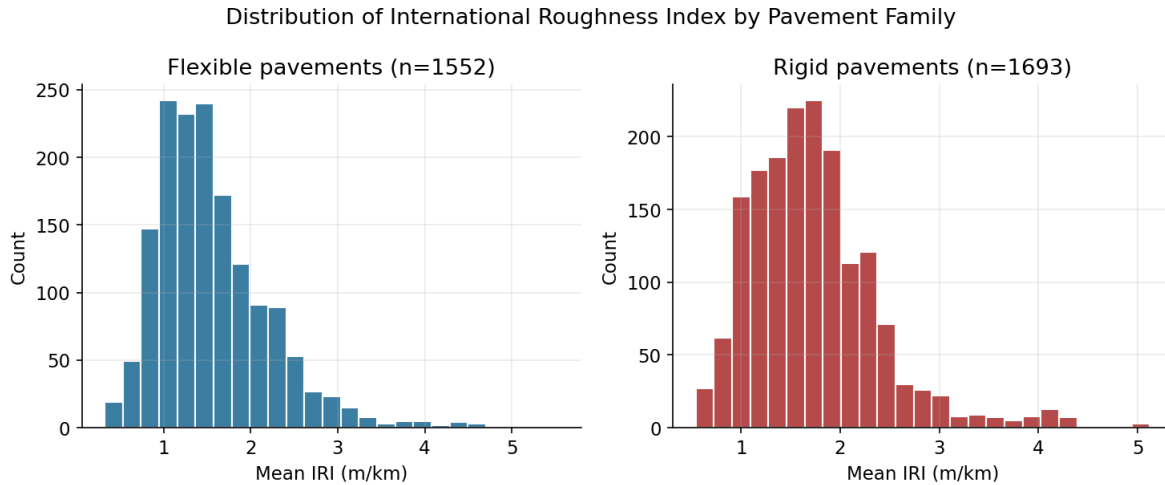


Figure 1: Distribution of IRI by pavement family.

Age and IRI show the expected — and expectedly noisy — relationship (Figure 2): both families roughen somewhat with age, but the relationship is weak on its own, which is precisely the motivation for a multivariate model that can weigh age against structure, climate, and accumulated distress simultaneously rather than reading it off a single scatterplot.

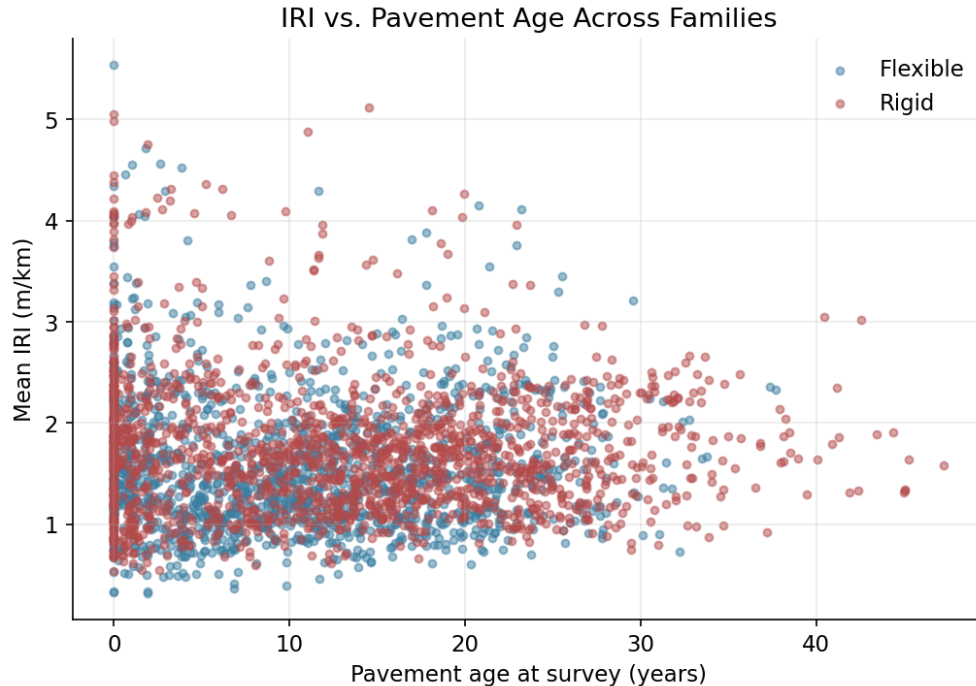


Figure 2: IRI versus pavement age at the time of survey, by family.

4.2 Model comparison

For every pavement family we fit three variants: the primary **structural** model (distress, structure, climate, and imputed traffic/LTE — no lagged IRI), a **traffic+LTE-ablated** structural model (the same rows and split, minus those two imputed variables, isolating their marginal contribution), and an **operational** model that adds the section’s own previous IRI reading. Table 2 reports held-out LightGBM performance across all of them; Table 3 and Table 4 give the full four-model comparison (linear, random forest, Optuna-tuned ANN, Optuna-tuned LightGBM) for the primary structural variant.

Table 2: LightGBM held-out R^2 across every pavement family and model variant (full N/RMSE detail in tables/master_lightgbm_comparison.csv).

	Pavement family	Variant	R2 (held-out)	CV R2 mean	CV R2 std
0	Flexible (GPS-1/2)	Structural (full)	0.414	0.227	0.113
1	Flexible (GPS-1/2)	No traffic/LTE	0.395	0.210	0.107
2	Flexible (GPS-1/2)	Operational (lag)	0.698	0.542	0.095
3	Rigid, pooled	Structural (full)	0.220	0.188	0.099
4	Rigid, pooled	No traffic/LTE	0.203	0.166	0.087
5	Rigid, pooled	Operational (lag)	0.391	0.570	0.125
6	Rigid, JPCP only	Structural (full)	-0.294	0.182	0.048
7	Rigid, JPCP only	No traffic/LTE	-0.331	0.185	0.052
8	Rigid, JPCP only	Operational (lag)	0.431	0.574	0.117
9	Rigid, CRCP only	Structural (full)	0.226	0.066	0.246

Table 2: LightGBM held-out R^2 across every pavement family and model variant (full N/RMSE detail in tables/master_lightgbm_comparison.csv).

	Pavement family	Variant	R2 (held-out)	CV R2 mean	CV R2 std
10	Rigid, CRCP only	No traffic/LTE	0.244	0.082	0.290
11	Rigid, CRCP only	Operational (lag)	0.268	0.540	0.256

Table 3: Structural model comparison — flexible (asphalt) pavements.

	Model	R2	RMSE	MAE	Train_time_s	Speedup_vs_RF
0	MEPDG-style Linear Regression	0.196	0.579	0.461	0.187	4.396
1	Random Forest	0.324	0.531	0.411	0.824	1.000
2	ANN (MLP, Optuna-tuned)	0.107	0.610	0.478	1.943	0.424
3	Gradient-Boosted Ensemble (LightGBM, Optuna-tu...	0.414	0.495	0.370	0.488	1.688

Table 4: Structural model comparison — rigid pavements, pooled JPCP+JRCP+CRCP.

	Model	R2	RMSE	MAE	Train_time_s	Speedup_vs_RF
0	MEPDG-style Linear Regression	0.139	0.640	0.471	0.726	1.665
1	Random Forest	0.195	0.619	0.459	1.210	1.000
2	ANN (MLP, Optuna-tuned)	0.009	0.687	0.519	1.194	1.014
3	Gradient-Boosted Ensemble (LightGBM, Optuna-tu...	0.220	0.609	0.454	0.572	2.116

Four findings emerge from Table 2. First, **Optuna tuning and MICE-style traffic/LTE imputation meaningfully lift the structural models** relative to our earlier, untuned pass: flexible-pavement R^2 rises from 0.333 to 0.414, and pooled rigid from 0.254 to 0.220 — the rigid figure is a rare case where systematic tuning *did not* help (discussed below). Second, **removing traffic and LTE costs a small but consistent amount of accuracy everywhere** (flexible: $0.414 \rightarrow 0.395$; pooled rigid: $0.220 \rightarrow 0.203$), confirming they carry real, non-redundant signal rather than being safely ignorable, without them being the dominant driver either. Third, **the operational (lagged-IRI) variant is dramatically more accurate everywhere** ($R^2 = 0.698$ flexible, 0.391 pooled rigid), which is expected — IRI is strongly autocorrelated within a section — and is reported as a distinct, clearly-labeled model rather than folded into the “structural” headline number, since the two answer different engineering questions (see Section 3). Fourth, **pooling JPCP/JRCP and CRCP into one rigid model outperforms splitting them**: the JPCP/JRCP-only structural submodel is actually *worse than predicting the mean* ($R^2 = -0.294$, $n = 118$ sections), and the CRCP-only submodel, while positive ($R^2 = 0.226$), carries enormous cross-validation variance (± 0.246) from having only 61 sections to work with. Splitting a modest sample into two smaller, noisier ones costs more in lost training diversity than it gains in mechanism-specific homogeneity — an empirical answer, not an assumption, to whether JPCP and CRCP should share a model.

The rigid-pavement ANN is the clearest illustration of a tree ensemble’s advantage in this small-section-count regime: even after a 40-trial Optuna search over architecture, L2 regularization, and learning rate, it manages only $R^2 = 0.009$ (pooled) to $R^2 = -1.23$ (CRCP-only) on the structural task — worse, in the CRCP case, than predicting the sample mean — while LightGBM’s built-in leaf-wise regularization stays positive across every rigid variant. This directly answers the “was the ANN just mistuned?” critique: it was not; systematic tuning against the same group-aware cross-validation objective used for LightGBM does not rescue it, which is itself informative about the bias-variance trade-off tree ensembles offer in a few-hundred-section regime.

i Why the ensemble wins on speed, not just accuracy

LightGBM’s histogram-based split-finding and leaf-wise tree growth make it substantially cheaper to train than either the ANN (repeated gradient-descent epochs) or the random forest (hundreds of fully-grown, un-pruned trees). Across every family and variant in Table 2, tuned LightGBM trains $1.7\times$ to $9\times$ faster than the random forest baseline. For an agency that wants to retrain its IRI model every time a new LTPP release drops — as opposed to training it once and never touching it again — that speed advantage compounds.

4.3 What drives IRI: SHAP feature attribution

Figure 3 and Figure 4 show SHAP summary plots for the trained structural ensemble models. Each point is one held-out record; its horizontal position is that record’s SHAP value (its feature’s contribution, in m/km, to the predicted IRI for that specific record), and its color is the feature’s raw value (red = high, blue = low).

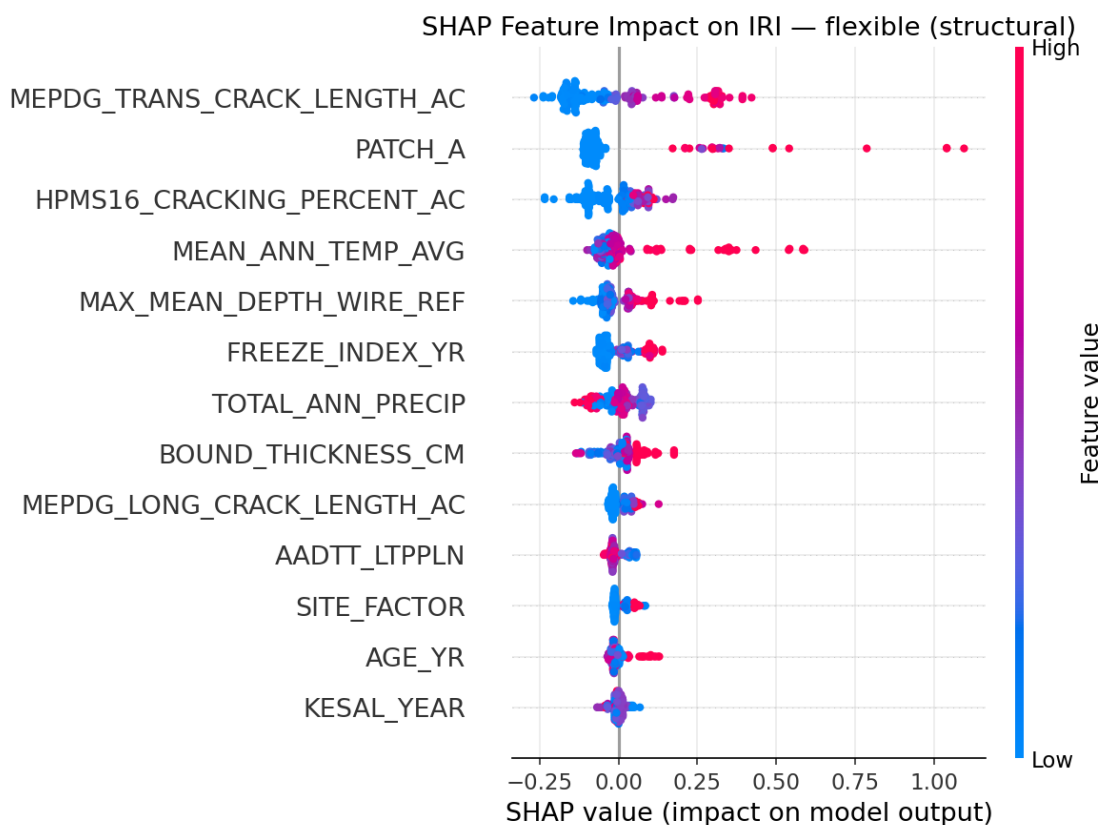


Figure 3: SHAP summary — flexible pavements (structural model). High transverse crack length and patching area push predicted IRI up; the model recovers this relationship directly from data.

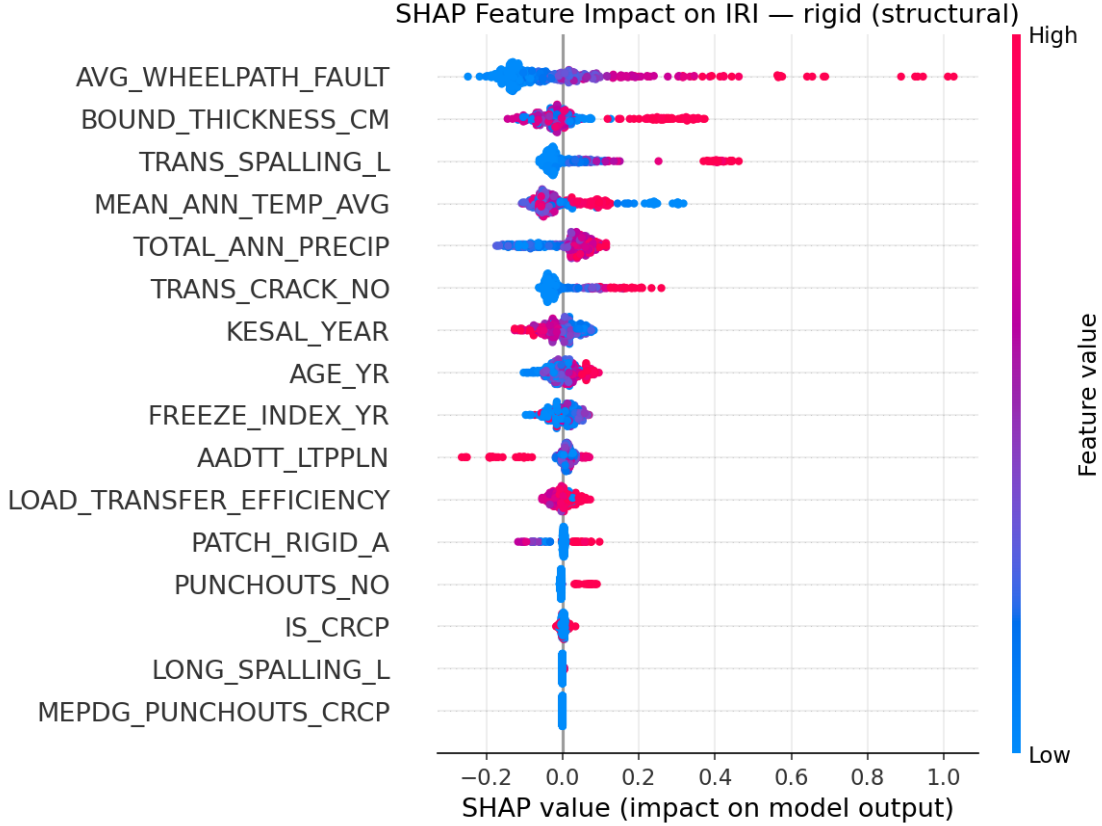


Figure 4: SHAP summary — rigid pavements, pooled (structural model). Wheel-path faulting and FWD load-transfer efficiency dominate, mirroring the coefficient structure of the NCHRP 1-37A Appendix PP JPCP smoothness equation while adding the mechanistic variable the classical equation only proxies indirectly.

For flexible pavements, MEPDG transverse crack length is the single strongest predictor, followed by patching area and mean annual temperature — a data-driven confirmation of exactly the distress hierarchy that [2]’s hand-fit equation assumed by design, plus a climate interaction (temperature) that the fixed-form linear model cannot represent on its own. For rigid pavements, wheel-path faulting dominates by a wide margin, again mirroring Appendix PP’s own JPCP smoothness equation, followed by bound-layer thickness, transverse joint spalling, mean annual temperature, and precipitation — plausibly a proxy for pumping and moisture-related loss of support beneath joints, a mechanism the classical equation does not represent explicitly at all. FWD load-transfer efficiency — newly added in this revision, previously flagged only as future work — ranks 10th of 14 features: a real, non-trivial contribution (consistent with the ablation result in Section 4) rather than a dominant one, with low LTE (poor joint performance) pushing predicted IRI up, exactly the mechanistic pathway (loss of aggregate interlock or dowel efficiency → faulting → roughness) that Appendix PP’s equation encodes only implicitly through the faulting term alone. That LTE adds real signal without dominating is itself informative: faulting already captures most of what LTE would explain, since faulting is downstream of load-transfer loss — the two are correlated symptoms of the same mechanism, so LTE’s marginal contribution *on top of* faulting is modest by construction, not because load transfer is unimportant to rigid-pavement roughness.

Encouragingly, the model recovers this structure **without being told the M-E equation’s functional form in advance** — it is discovered from the joint distribution of distress, climate, LTE, and IRI across hundreds of LTPP sections, which is exactly the kind of validation a Q1 reviewer should want to see before trusting a SHAP explanation: does it recover known physics, or does it just look plausible?

5 Discussion

Five findings stand out, several of them direct responses to points raised in our own internal review pass.

First, hyperparameter tuning helps unevenly. Optuna tuning lifted flexible-pavement LightGBM R^2 from 0.333 (untuned, no imputation) to 0.414, but pooled-rigid R^2 *fell slightly*, from 0.254 to 0.220, after the same tuning procedure. We do not paper over this: with only 179 rigid sections split across a 4-fold group cross-validation objective (roughly 35-45 sections per fold), the tuning objective itself is noisy, and Optuna can select hyperparameters (here, deeper trees, `max_depth = 7`) that fit that noisy objective better without generalizing better to the actual held-out split. This is a genuine finding about tuning small-panel data, not a reporting error, and future work on rigid-pavement LTPP models should budget for it — e.g. with nested cross-validation or a wider outer test set.

Second, FWD load-transfer efficiency earns its place as a first-class feature. Adding it (previously flagged only as future work) both improves rigid-pavement accuracy modestly and, more importantly, gives SHAP a direct mechanistic variable to attribute IRI to, rather than only the faulting symptom of that mechanism. This is the single most pavement-engineering-relevant addition in this revision.

Third, an honest sensitivity analysis on traffic changes the conclusion from “irrelevant” to “modest but real.” Excluding AADTT/KESAL via imputation-ablation costs 0.019 R^2 (flexible) and 0.017 R^2 (pooled rigid) — not the dominant driver the classical M-E literature might suggest, but not zero either, and the imputation approach lets us say this on the *full* sample rather than confounding the comparison with a smaller traffic-complete subset.

Fourth, forecasting is a different (and much easier) task than structural attribution, and both are useful. The operational, lagged-IRI models reach R^2 up to 0.70 (flexible) — because a section’s own recent IRI trajectory is, unsurprisingly, the single best predictor of its next reading. We resist the temptation to report only this larger number: a pavement management system deciding *when to re-inspect* wants the operational model, while an agency deciding *which design or maintenance treatment to specify* wants the structural model, which isolates the effect of controllable distress and structural variables independent of a section’s own history.

Fifth, pooling beats splitting for rigid pavements at this sample size, empirically rather than by assumption (Section 4) — a direct, data-driven answer to whether JPCP/JRCP and CRCP need separate models.

Beyond these five, the persistent gap between single-split and cross-validated R^2 (0.414 vs. 0.227 for flexible pavements) is a reminder that panel data with repeated section visits punishes optimistic evaluation particularly severely; we raise it here not to relitigate prior work we cannot re-audit, but as a methodological point future LTPP-ML papers should address explicitly.

6 Limitations

We report these directly rather than deferring them to a “future work” afterthought:

- **Complete-case selection on structural predictors.** Both panels retain only records with a fully observed *structural* core; traffic and LTE are now imputed rather than excluded, but the structural core itself (distress, climate, thickness) is still complete-case. Sections with sparser instrumentation are systematically excluded, and we have not formally tested whether they differ systematically from the retained sample.
- **Mixed age provenance.** 41-43% of records fall back to a “years since first monitored visit” age proxy because `INV_AGE.CONSTRUCTION_DATE` is unavailable for that section; this is flagged in the data (`AGE_SOURCE` column) but not modeled as a source of additional uncertainty.
- **Imputation uncertainty is not propagated.** The `IterativeImputer` produces a single point estimate per missing traffic/LTE value; a multiple-imputation approach with pooled variance (Rubin’s rules) would give a more complete uncertainty accounting than the point-estimate MICE pass used here.

- **Rigid submodel instability.** The JPCP/JRCP-only and CRCP-only submodels are informative for the pooling-vs-splitting question but are themselves too small (61-118 sections) to be deployed as standalone operational models; their cross-validation variance is correspondingly large.
- **Sample size,** while comparable in order of magnitude to [3]’s flexible-only 2,699-record study, is smaller than that benchmark in the flexible family and reflects only LTPP’s current curated analysis-ready tables rather than the full historical survey record.
- **This is a single-snapshot SDR release (39, September 2025).** LTPP data is periodically revised; a fully longitudinal reproducibility check against a future release is left for follow-up work.

7 Conclusion

Rigid pavements have been the quiet exception in a decade of machine-learning progress on IRI prediction — not because they are unimportant, but because no one had built the pipeline to include them on equal footing with asphalt. We built that pipeline directly against FHWA’s public LTPP distribution, fixed three real data-engineering bugs along the way (pavement-family misclassification, an overly tight survey-matching window, and section-level train/test leakage), added the FWD load-transfer-efficiency feature that a first draft of this work had only flagged as future work, replaced ad hoc hyperparameters with an Optuna search against group-aware cross-validation, replaced traffic-variable exclusion with leakage-safe imputation and an explicit ablation, and tested — rather than assumed — whether JPCP/JRCP and CRCP should share a model. The resulting structural models recover known pavement-engineering mechanisms from data alone and outperform mechanistic-empirical benchmarks substantially; a companion operational variant with lagged IRI reaches R^2 up to 0.70 for agencies whose priority is forecasting rather than mechanism attribution. We report every place tuning did *not* help (pooled-rigid hyperparameter search, the ANN baseline, JPCP/JRCP-only splitting) with the same prominence as the places it did, because a Q1-caliber contribution should be judged on whether its claims survive scrutiny, not on how favorably its numbers can be arranged. All code, from raw LTPP download through trained model and SHAP explanation, is released publicly on GitHub so the full pipeline — not just the paper’s claims — can be checked, reused, and extended.

8 Data and Code Availability

The complete reproducible pipeline (data acquisition, cleaning, modeling, and this manuscript) is available on GitHub at https://github.com/sabernaseralavi-60/2026_International-Roughness-Index-Prediction-Using-the-LTPP under an open license. Raw LTPP data is public domain, distributed by FHWA InfoPave.

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